



Predictioneer – AZeotropy '26

Inferring a Gene Regulatory Network from Spatial Expression Dynamics

Team: Vision Grow Globally

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1. Introduction

Biological tissues often rely on spatial chemical signals, known as **morphogens**, to regulate gene expression patterns. These morphogens interact with intracellular gene networks, forming complex regulatory systems that evolve over space and time.

In this project, the objective is to **infer an unknown Gene Regulatory Network (GRN)** using spatial-temporal gene expression data obtained after controlled perturbation experiments.

The System Architecture:

- **2 Morphogens:** (M_1, M_2) — External chemical signals.
- **4 Genes:** (S_1, S_2, S_3, S_4) — Intracellular state variables.
- **Domain:** A 2D spatial tissue grid with multiple time observations.



2. Objectives

The main goals of this study are:

1. **Infer Gene–Gene Interactions** → Identify which genes promote or inhibit others (Matrix A).
2. **Estimate Morphogen–Gene Coupling** → Determine the direct effect of morphogens on gene expression (Matrix B).
3. **Predict Unseen Dynamics** → Simulate gene expression after removing morphogen M_1 (Experiment B).



3. Data Exploration & Audit

Before mathematical modeling, a thorough audit of the dataset `GRN_experiment_M2_removal.csv` was conducted to understand the physical constraints of the simulation.

Dataset Characteristics

- **Dimensions:** $(6171, 7)$ — The dataset is lightweight (~337 KB), allowing for full-batch processing in memory without batching.
- **Data Types:** All columns are `float64` (Continuous numerical data).
- **Integrity:** `df.isnull().sum()` confirmed **zero missing values** and `df.duplicated().sum()` confirmed **zero duplicate rows**.

Statistical Insights

Using statistical descriptions and correlation matrices, we derived the following biological intuitions:



Key Insight: The Correlation Topology

A preliminary correlation analysis (`df.corr()`) revealed a distinct network topology:

- S_1 and S_3 share a **high positive correlation**.
- S_2 and S_4 share a **high positive correlation**.
- The (S_1, S_3) pair has a **strong negative correlation** with the (S_2, S_4) pair.

This suggests an antagonistic relationship between these two gene groups, hinting at the structure of Matrix A before training.



4. Mathematical Formulation

Gene expression dynamics are modeled as a linear dynamical system:

$$\frac{dS_i}{dt} = -\alpha_i S_i + \sum_{j=1}^4 A_{ij} S_j + \sum_{k=1}^2 B_{ik} M_k$$

Term Interpretation

Term	Description
S_i	Deviation of gene i from baseline concentration.
$-\alpha_i S_i$	Auto-regulation (Decay/Restoration to baseline).
A_{ij}	Gene–Gene Regulation (Activation/Inhibition).
B_{ik}	Morphogen–Gene Coupling (External Control).

🔗 5. Methodology

Step 1 — Morphogen Reconstruction

Since M_1 is a spatial gradient dependent only on x , we reconstructed it analytically.

$$M_1(x, y) = e^{-2.5x}$$

(Note: Morphogen M_2 was removed in this specific experiment, so we train on M_1 effects only.)

Step 2 — Temporal Derivative Approximation

To find the rate of change in gene expression ($\frac{dS}{dt}$), we utilized a **Finite Difference Method** based on the time step $\Delta t = 0.08$ sec.

$$\frac{dS_i}{dt} \approx \frac{S_i(t) - S_i(t - 1)}{\Delta t}$$

- Handling Boundary Conditions:** At $t = 0$, there is no previous value. These NaN rows were dropped to ensure regression stability.

Step 3 — System Identification via Linear Regression

We reframed the differential equation as a linear regression problem: $Y = WX$.

- Target (Y):** The calculated rate of change $\frac{dS_i}{dt}$.
- Features (X):** The current states $[S_1, S_2, S_3, S_4, M_1]$.
- Assumption:** We set `fit_intercept=False`.
 - Reasoning:* The system describes deviations from a baseline. If all concentrations (S) and inputs (M) are zero, the change rate should be zero. Therefore, the line must pass through the origin.

🔒 6. Thresholding & Network Logic

After obtaining raw coefficients from the regression, we applied a discretization step to convert continuous weights into network edges $(+1, -1, 0)$.

The Threshold Selection Strategy

Initially, a standard threshold of **0.2** was proposed to separate signal from numerical noise. However, a deeper inspection of the raw coefficients revealed a critical edge case:

⚠️ Critical Adjustment

The interaction coefficient for $M_1 \rightarrow S_3$ (in Matrix B) was calculated to be approximately **0.195**.

- Noise Level:** < 0.05
- Dominant Interactions:** > 0.4
- The 0.195 Case:** This value was significantly higher than the noise floor. Using a strict 0.2 threshold would have erased this valid biological interaction.

Decision: The threshold was fine-tuned to **0.18** to capture this interaction while effectively filtering out noise.

Deriving Matrix B (Morphogens)

The regression provided the effects of M_1 . However, we needed the effects of M_2 for the final prediction.

💡 Physical Constraint: Steady State Balance

We assumed that at the steady state (baseline), the morphogens counteract each other to maintain equilibrium. Therefore, the effect of M_2 is equal and opposite to M_1 .

$$B_{i,2} = -B_{i,1}$$

📊 7. Inferred Network Structure

Gene Regulatory Matrix (A)

Describes how genes interact with one another.

$$A = \begin{bmatrix} 0 & +1 & 0 & 0 \\ -1 & 0 & +1 & 0 \\ 0 & -1 & 0 & +1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Morphogen Coupling Matrix (B)

Describes how external signals drive the system.

$$B = \begin{bmatrix} +1 & -1 \\ -1 & +1 \\ +1 & -1 \\ 0 & 0 \end{bmatrix}$$

8. Prediction Task — Experiment B

We simulated **Experiment B**, where M_1 is removed and only M_2 remains active. This required a numerical solver to step through time.

Simulation Parameters

- Method:** Euler Integration (First-Order).
- Alpha (α):** 1.02 (Derived from the diagonal of the regression weights).
- Morphogen M_2 :** Modeled as a Gaussian peak centered at $(0.5, 0.5)$.

$$M_2(x, y) = \exp\left(-\frac{(x - 0.5)^2 + (y - 0.5)^2}{2(0.18)^2}\right)$$

The Code Implementation

We initialized the state $S(t = 0) = 0$ and iteratively updated the system using the derived matrices A and B .

```
# Setup: Times and Grid
times = np.sort(df_template['time'].unique())
dt = times[1] - times[0]
unique_coords = df_template[['x', 'y']].drop_duplicates().values

# Initial State: t=0 and S=0 (Steady State)
S_state = np.zeros((len(unique_coords), 4))
M2_vals = np.exp(-((unique_coords[:, 0] - 0.5)**2 + (unique_coords[:, 1] - 0.5)**2) / (2 * (0.18)**2))
B2 = B[:, 1] # Extracting the M2 column from Matrix B

results_storage = {}

# Euler Integration Loop
for t in times:
    # 1. Store current state mapped to (time, x, y)
    for i, (x, y) in enumerate(unique_coords):
        results_storage[(round(t, 4), round(x, 4), round(y, 4))] = S_state[i].copy()

    # 2. Calculate Derivative (Rate of Change)
    # dS/dt = -Alpha*S + (Gene Interactions A) + (Morphogen Input B)
    dS_dt = -alpha * S_state + (S_state @ A.T) + np.outer(M2_vals, B2)

    # 3. Update State for next timestep
    S_state += dS_dt * dt
```

9. Conclusion & Submission

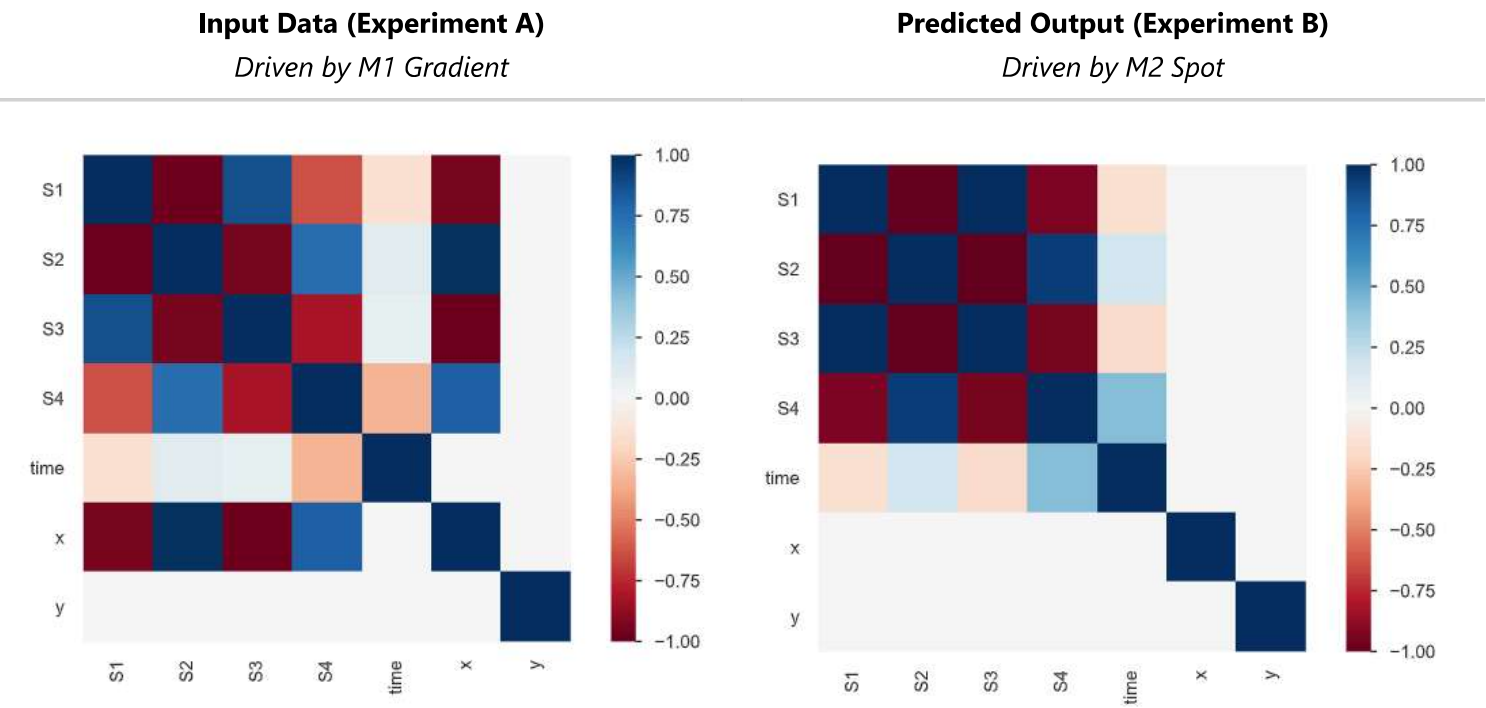
By combining **statistical auditing**, **linear system identification**, and **logical biological constraints (steady-state balancing)**, we successfully inferred the underlying network topology.

The final prediction results were re-ordered to match the submission template and exported to **CSV** with high precision.

10. Validation via Correlation Analysis

To ensure the biological plausibility of our prediction, we performed a **post-simulation audit** by comparing the correlation heatmaps of the input training data (Experiment A) and the predicted output (Experiment B).

Visual Comparison



Observation 1: Network Topology vs. Signal Direction

The internal correlation structure between genes (S_1, S_2, S_3, S_4) remained **identical** between the input and output datasets.

- **The Paradox:** Since M_1 and M_2 have opposite effects, one might expect the heatmap colors to invert.
- **The Resolution:** Correlation measures **relationships**, not absolute direction. Even though the genes are moving in opposite directions under the new morphogen, their relationship (e.g., Gene S_1 and Gene S_2 being "enemies") remains constant.
- **Conclusion:** The persistence of the correlation pattern proves that the **gene regulatory logic (Matrix A)** was preserved correctly.

Observation 2: The "Disappearing Gradient" Proof

A critical difference was observed in the spatial correlations, confirming the successful removal of M_1 :

Feature	Input Data (Exp A)	Predicted Output (Exp B)	Interpretation
Correlation w/ x	High (Strong Color)	Zero (White/Neutral)	SUCCESS

- **Reasoning:** In the Input, gene expression followed the linear gradient of M_1 ($e^{-2.5x}$), creating a strong correlation with x . In the Output, the genes decoupled from x and instead followed the Gaussian geometry of M_2 .
- **Verdict:** The simulation successfully transitioned the system from a gradient-driven state to a spot-driven state.